Enthalpy and Free Energies If you drop an aluminum ball into a drain cleaner solution; it is already in **consta**

Now if you burn hydrogen gas inside a tube filled chlorine gas, you will observe the light blue flame, and hydrochloric gas is
[width=0.4] /PhyAI/chapter4/$4_1.png$ How much energy required to create steam from out of nothing? The answer is quite

Like entropy, we actually do not care much about the enthalpy $H$ of a stable system; it is not so meaningful when I say "A

I have always thought of enthalpy as a representation of the state of all substances, in terms of energy. Look on the left side

Conversely, the surrounding environment needs to **provide** energy to decompose $H2O$ into $H2$ and $O2$; this is called an **en**

The enthalpy of vaporization of water at 100°C is 40.63 (kJ/mol). How much energy is needed to create space for steam?
From Definition , we see that:

The amount of work $PV$ can be caculated by using ideal gas law:

Under **constant pressure** condition, by performing some mathematical transformations, we obtain: $\Delta H = \Delta U + P\Delta V$
$= Q + W_{other}$ Have you notice a new quantity $W_{other}$? Unlike us, chemists have many ways to supply work to system with

Gibbs Free Energy If you take into account the influence of the surrounding environment on the steam rising experiment, yo

[H]
[width=0.6] /PhyAI/chapter4/$4_2.png$ Definition of Gibbs free energy [Gibbs Free Energy Definition] The Gibbs free energy

Like enthalpy $H$, we do not really care about the total Gibbs free energy $G$ of a system rather than the difference $\Delta G$ for a

We can find the values of $H$ and $S$ of any substances in the table "Thermodynamics Properties" at 298K and 1 bar. But ho
If $\Delta H < 0$ and $\Delta S > 0$, that means $\Delta G < 0$, the reaction is always spontaneous at all temperature.
If $\Delta H < 0$ and $\Delta S < 0$, that means the reaction is only spontaneous at low temperature.
If $\Delta H > 0$ and $\Delta S < 0$, that means $\Delta G > 0$, the reaction is never spontaneous at all temperature.
If $\Delta H > 0$ and $\Delta S > 0$, that means the reaction is only spontaneous at high temperature.
Let's look carefully on our quantities. From definitions above, we already know that both $\Delta H$ and $\Delta G$ indicate how much

Everyone knows that $\Delta S < 0$ (water is more "ordered" than gases), so can we conclude this reaction is not spontaneous? A

Wow, so even $\Delta S < 0$, this reaction is still **spontaneous** at normal room condition, because $\Delta G < 0$. Observing the proces
$T\Delta S = -13.23 \quad (kJ)$
The amount of energy needed for decreasing entropy is just 13.23kJ, but the reaction releases 241.82kJ to surrounding; so w

[Helmholtz Free Energy Definition] The Helmholtz free energy is defined as:

We do not have much to discuss Helmholtz energy here, because in fact, it is much easier to conduct our experiment at cons
[width=0.5] /PhyAI/chapter4/$4_3.png$ "Uniform Mercury" Legendre Transformation By applying the thermodynamics iden
=(TdS-PdV+$\mu dN) + PdV + VdP$
$= TdS + \mu dN + VdP$ This is an example of **Legendre transformation**; there are many ways you can do transformation
=(TdS-PdV+$\mu dN) - TdS - SdT$
$= -PdV + \mu dN - SdT$

Similarly, for Gibbs free energy: dG=dF+PdV+VdP=$\mu dN - SdT + VdP$

These 2 relationships are so important, they verify our intuition is correct: we have to provide more external energy to decr